POLARIN

POLAR RESEARCH INFRASTRUCTURE NETWORK

Technical documentation

Architecture Design Document

Versions

Version	Date	Comment	Responsible
0.2	2024-11-27	added more details to most sections	Daan Kivits
0.1	2024-09-18	Outline modified from SIOS ADD.	Øystein Godøy

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1. Introduction

1.1. Background

POLARIN is an international network of polar research infrastructures and their services, aiming at addressing the scientific challenges of the polar regions. The network includes a wide array of complementary and interdisciplinary top level research infrastructures: Arctic and Antarctic research stations, research vessels and icebreakers operating at both poles, observatories, data infrastructures and ice and sediment core repositories.

POLARIN will provide integrated, challenge-driven, and combined access to these infrastructures to facilitate interdisciplinary research on complex processes. POLARIN has the following six specific objectives, coherent with the work packages, each with measurable outcomes:

1. Enable science for understanding and predicting key processes in polar regions.
2. Provide efficient challenge-driven transnational access (TA) to top level RIs in the polar regions.
3. Improve data services and provide customised data products.
4. Provide virtual access (VA) to data and data services.
5. Provide training for infrastructure users.
6. Advertise RI services and engage RI users.

This document is addressing user requirements for items 3 and 4 above, with specific emphasis on item 4 - providing virtual access to data. More information on the POLARIN project can be found in the Description of Work described in the POLARIN project proposal RD-1.

NOTE	The first version of this document was developed from the a similar Use Case document used within Svalbard Integrated Arctic Earth Observing System (SIOS) (RD-2).

1.2. Definitions

The term *data product* is quite wide and in the remaining the term *dataset* will be used. This can be observational data but also processed observations etc for a specific purpose. The working definition of a dataset is in line with the INSPIRE directive:

> an identifiable collection of spatial data, i.e. a collection of data that has a reference by name or coordinates to a geographic location or area, and which in addition have a designated start and end time.

A dataset can contain observations (remote or in situ), derived quantities (from either of these two types of data sources) or forecasts of future states of environmental parameters. Data values can be located at a single point, along a line or transect, in a regular or irregular grid, and be captured or estimated at one or more altitudes/depths. A dataset can be stored on paper, in files (one or more), or in a database, and is often accompanied by descriptions (metadata) of its content.

1.3. Scope of document

This document describes the architecture of POLARIN Virtual Access. It is not an official deliverable of the project, but aims to establish a common foundation for the work in work packages 4 and 5.

NOTE This document is not a formalised Architecture Document following a formalised approach like RM-ODP (Reference Model for Open Distributed Processing), but rather a lightweight architecture approach.

1.4. Intended audience

Data and system managers within POLARIN, in particular participants in POLARIN Work Packages 4 and 5, "Improvement of data services and customised data products" and "Provision of virtual access" respectively.

1.5. Applicable documents

RD-1 Description of the action (DOA) (Associated with document Ref. Ares(2023)7293125 - 26/10/2023)

RD-2 SIOS Data Management System Architecture Document

RD-3 POLARIN Web Portal Use Case Document (in draft version)

RD-4 Concept document (not available yet)

2. Overall concept

An outline of the division of tasks between work package 4 and 5 is provided in Figure 1, and a more detailed description can be found in Section 3.1. This diagram indicates that development of the human front end to the POLARIN data catalogue is done within WP 4, while virtual access as such is provided in WP 5. The latter is done through compilation of the POLARIN data catalogue in WP 5 (Section 3.2), which is then utilised by the human front end developed in WP 4 (Section 3.3). The requirements that the human interface has to fulfil are outlined in RD-3.

NOTE We should use a more simple, conceptual architecture diagram here, similar to Figure 1 but less detailed.

3. Detailed architecture

This chapter describes the overall architecture of the proposed POLARIN web portal, worked on in both WP 4 and WP 5. Section 3.2 describes the design of the POLARIN data catalogue (back-end), and Section 3.3 describes the design of the human readable interface (front-end) that exposes the metadata from the data catalogue.

3.1. Architecture overview

NOTE Figure 1 should be updated according to the input from WP 4 regarding the implementation of the human front-end (described in more detail in Section 3.3)

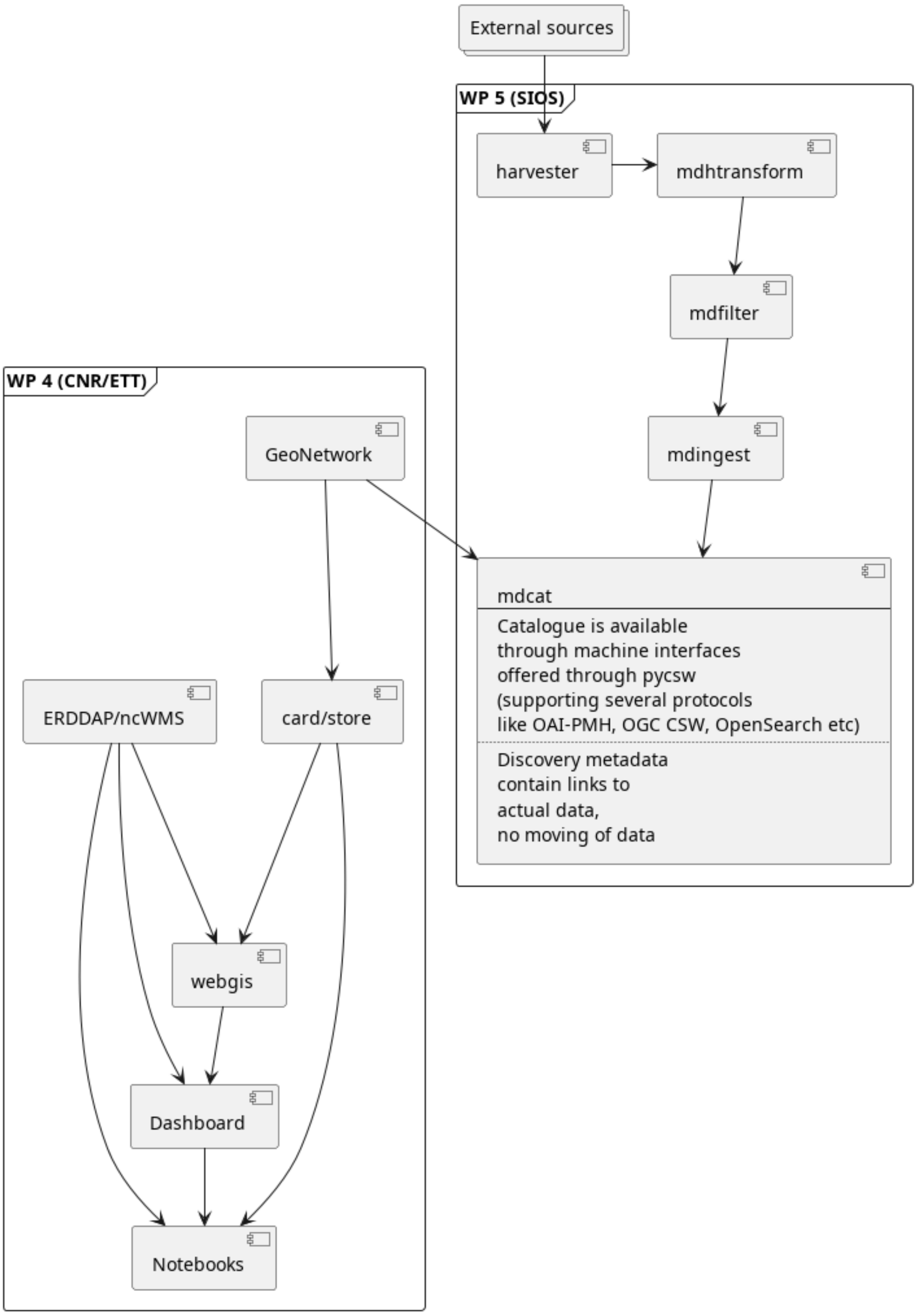

Figure 1. Architecture diagram

The datasets relevant for POLARIN may be hosted in a number of data centres. Some of these are members of POLARIN, while others are not but have been used by research infrastructures/stations for a long period. All datasets have to be properly tagged with POLARIN as a project name. This information is used to filter out the relevant datasets for POLARIN.

NOTE A minimum discovery metadata profile has been identified and is further described below.

3.2. Data catalogue

The proposed POLARIN data catalogue as designed in WP 5 will adhere to the structure given in Figure 1, according to the following pipeline: metadata records are harvested from multiple external resources; the metadata records are then translated into an internal metadata model; the translated metadata records relevant to POLARIN are identified and filtered; the filtered metadata records are then ingested into a central database / search engine and then exposed in a standard machine-readable interface.

Here is a short overview of the different parts of the data catalogue setup:

(md)harvester
: this is where all discovery metadata records from the data centers ('external sources' in the schematic) are harvested. This will be the connection to standardised machine-readable endpoints.

mdtransform
: this is where the metadata standards provided by external resources (i.e. the metadata harvested in the previous step) are translated to the internal metadata model.

mdfilter
: this is where POLARIN relevant metadata is filtered.

mdingest
: this is where the filtered metadata records are ingested into SolR (search engine). SolR is the backend of the mdcat (machine readable interface – pyCSW).

mdcat
: this is the machine-readable interface of the data catalogue, that offers machine-to-machine (M2M) protocols through a pyCSW implementation. This includes (but is not limited to) OAI-PMH, OGC CSW, and OpenSearch.

The flexible architecture of pyCSW allows for the relatively efficient implementation emerging M2M protocols, making it future-proof for evolving standards and specialized needs. Overall, pyCSW's support for multi-protocol, machine-readable interfaces strengthens data accessibility and fosters collaboration, ensuring continued relevance as data-sharing standards evolve.

3.3. Human frontend

NOTE The front end can add additional external sources to the human front end, but will have to crosscheck for uniqueness with the datasets using the identifiers of the datasets found in the POLARIN data catalogue.

FIXME

4. Documentation standards

4.1. Minimum discovery metadata profile

A list of generic elements is provided in Table 1, together with a high level explanation. In Table 2 a suggested mapping between the metadata elements and commonly used metadata standards is provided.

NOTE It is expected that a number of metadata standards has to be supported, including various profiles of ISO-19115, schema.org (preferably ESIP's science on schema.org), GCMD DIF, STAC etc.

Table 1. A minimum discovery metadata profile has been defined.

Element	Purpose
Metadata identifier	A unique identifier for the dataset. This is used to avoid duplicate records in aggregator catalogues. Utilisation of UUID with a namespace prefix is recommended.
Last update of metadata	An ISO8601 datetime for the last update of the metadata record.
Title	To provide a brief explanatory title for the dataset
Abstract	A short summary of the dataset, its purpose and how it was generated.
Temporal Extent	The temporal spanning of the dataset. This can be a multi temporal dataset.
Geographical Extent	The geographical location of the dataset. This can be a point, a bounding box, trajectory or a polygon.
Keywords	Keywords describing the dataset. Ideally this comes from controlled vocabularies and describes the variables of the dataset.
Personnel	Identification of all people that have contributed to the dataset. This requires full name, email, affiliation and a role description in the dataset. The latter has to come from a controlled vocabulary.
Publisher	This is identification of the data centre publishing the data. It contains a long and short name for the data centre and URL to the landing page.
Use constraint	This is a license for the data. This is a URL to the license text and an identifier. Utilisation of SPDX is recommended.
Data Access	This provides direct access to the dataset for download etc. It is not a landing page, but a direct link to the data and indication using a controlled vocabulary of the access mechanism (ranging from direct download to OPeNDAP and OGC WMS).
Project	A list of projects that has contributed to the creation of the dataset. Polarin has to be one of the projects.

Table 2. A suggested mapping between the elements of the minimum discovery metadata profile and some commonly used metadata standards.

Metadata field (Table 1)	**ISO-19115 (ISO-19139, ISO-19115)**	**DIF 10.2 (DIF-10.2)**	**Schema.org (Schema.org)**	**DCAT (DCAT-2.0)**	**ACDD (ACDD)**
Metadata identifier	gmd:fileIdentifier	Entry_ID	identifier	dct:identifier	id
Last update of metadata	gmd:dateStamp	Metadata_Dates/Metadata_Last_Revision	dateModified	dct:modified	date_metadata_modified
Title	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:citation/gmd:CI_Citation/gmd:title	Entry_Title	name	dct:title	title
Abstract	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:abstract	Summary/Abstract	description	dct:description	summary
Temporal Extent	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:extent/gmd:EX_Extent/gmd:temporalElement/gmd:EX_TemporalExtent/gmd:extent/gml:TimePeriod/gml:beginPosition (gml:endPosition)	Temporal_Coverage/Range_DateTime/Beginning_Date_Time (Ending_Date_Time)	temporalCoverage	dct:temporal	time_coverage_start and time_coverage_end
Geographical Extent	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:extent/gmd:EX_Extent/gmd:geographicElement/gmd:EX_GeographicBoundingBox/gmd:westBoundLongitude (gmd:eastBoundLongitude, gmd:southBoundLatitude, gmd:northBoundLatitude)	Spatial_Coverage/Geometry/Bounding_Rectangle/Westernmost_Longitude (Northernmost_Latitude, Southernmost_Latitude, Easternmost_Longitude)	spatialCoverage	dct:spatial	geospatial_lat_min,geospatial_lat_max, geospatial_lon_min,geospatial_lon_max

Metadata field (Table 1)	ISO-19115 (ISO-19139, ISO-19115)	DIF 10.2 (DIF-10.2)	Schema.org (Schema.org)	DCAT (DCAT-2.0)	ACDD (ACDD)
Keywords	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:descriptiveKeywords/gmd:MD_Keywords/gmd:keyword (and gmd:identificationInfo/gmd:MD_DataIdentification/gmd:descriptiveKeywords/gmd:MD_Keywords/gmd:thesaurusName/gmd:CI_Citation/gmd:title)	Science_Keywords for GCMD Science Keywords (or Ancillary_Keyword for free text)	keywords	dcat:theme (or dcat:keyword for free text)	keywords, keywords_vocabulary
Personnel	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:pointOfContact/gmd:CI_ResponsibleParty/gmd:individualName (gmd:organisationName, gmd:contactInfo and gmd:role)	Personnel/Role and Personnel/Contact_Person/First_Name (Last_Name, Email)	creator, contributor	dcat:contactPoint, dct:creator	creator_email, creator_institution, creator_name, creator_type, creator_url, contributor_name, contributor_role
Publisher	gmd:distributionInfo/gmd:MD_Distribution/gmd:distributor/gmd:MD_Distributor/gmd:distributorContact	Organization/Organization_Name/Short_Name (Long_Name) and Organization/Organization_URL	publisher	dct:publisher	publisher_name, publisher_email, publisher_institution, publisher_type, publisher_url
Use constraint	gmd:identificationInfo/gmd:MD_DataIdentification/gmd:resourceConstraints/gmd:MD_LegalConstraints/gmd:useConstraints (type=otherRestrictions) and gmd:identificationInfo/gmd:MD_DataIdentification/gmd:resourceConstraints/gmd:MD_LegalConstraints/gmd:otherConstraints	Use_Constraints	license	dct:license	license

Metadata field (Table 1)	ISO-19115 (ISO-19139, ISO-19115)	DIF 10.2 (DIF-10.2)	Schema.org (Schema.org)	DCAT (DCAT-2.0)	ACDD (ACDD)
Data Access	gmd:distributionInfo/gmd:MD_Distribution/gmd:transferOptions/gmd:MD_DigitalTransferOptions/gmd:onLine/gmd:CI_OnlineResource/gmd:linkage (and gmd:protocol)	Related_URL/URL_Content_Type/Type and Related_URL/URL	Distribution and Distribution/contentUrl	dcat:distribution (dcat:downloadURL and dcat:accessService)	TBD
Project	can be providede through keywords using "project" as gmd:MD_KeywordTypeCode	Project/Short_Name (and Long_Name)	funding	TBD	project

4.2. Data documentation standards

POLARIN is recommending usage of CF-NetCDF and Darwin Core Archives (DwC-A) for documentation of datasets. These are internationally recognized file formats that allow for providing embedded metadata in the file structure itself. By making datasets self-describing, these formats significantly reduce the need for human intervention in the data life cycle, from ingestion to reuse.

These file formats also align with ENVRI-FAIR recommendations for FAIR (Findable, Accessible, Interoperable, Reusable) data practices.

CF-NetCDF

NetCDF is a binary format that is most suitable for storing multi-dimensional array-based numerical scientific data, and can be made FAIR-compliant when combined with the adoption of the Climate and Forecast (CF) and Attribute Convention for Data Discovery (ACDD) conventions. The file format is recommended for meteorological, oceanographic, hydrological and glaciological data, but other types of data might also be suitable for CF-NetCDF.

Darwin Core Archive

Darwin Core is a data standard originally developed for biodiversity informatics, though this has expanded to be useful for various types of data associated with one or a list of organisms. It is the backbone of the Global Biodiversity Information Facility GIBIF[1] and the Ocean Biodiversity Information System (OBIS[2]). It consists of a set of comma separated files (CSV) bundled with a metafile (meta.xml) and using controlled vocabularies to describe the content. A second file (eml.xml) describes the content and strucuture of the CSVs and links the column headers to the Darwin Core terms.

4.2.1. Data granularity

POLARIN recommends to follow the following general guidelines on dataset granularity:

1. as a general guideline, always publish data at the highest possible functional granularity (i.e. not individual measurements, but neither several stations combined in one dataset)
2. never combine data with different temporal dimensions (e.g. minute and hourly resolutions) in the same dataset.
3. never combine data with different vertical dimensions (e.g. surface observations and vertical profiles) in the same dataset.
4. if there is a need to reference datasets to collection work (e.g. research cruises or field work), establish this reference through tags on the data or parent/child relations allowing data consumers to collect data based on content and spatio-temporal position.

These guidelines are based on the granularity perspectives practised by the Svalbard Integrated Arctic Earth Observing System (SIOS), which are explained in more detail here.

4.2.2. Types of metadata and machine interoperability

There exists several types of metadata, which serve different purposes. The most relevant in this context are:

Discovery metadata

This type of metadata describes who did what, where and when, how to access data and potential constraints on the data. It shall also link to further information on the data, if relevant. It includes information like titles, keywords, abstracts, geographic locations, dates, and creators. Its primary role is to make resources discoverable in catalogs, repositories, or search engines.

NOTE Examples of discovery metadata: ISO-19115, GCMD DIF and ACDD.

Use metadata

This type of metadata describes the actual contentof a dataset and how it is encoded. The purpose is to enable the user to understand the data without any further communication. It describes content of variables using standardised vocabularies, units of variable, encoding of missing values, map projections etc. Use metadata is especially important to build and provide services and tools on top of the data, as it ensures that users and machine interfaces can effectively work with the resource once it is found.

NOTE Examples of use metadata are: Climate and Forecast (CF convention) and Darwin Core Archive

The use of standardized metadata enables the use of M2M interfaces, which are vital for automating data discovery, integration, and analysis. This is key to support the harvesting procedures of POLARIN, enhancing usability and enabling complex, automated workflows across various scientific domains.

4.2.3. Supporting data interoperability across scientific domains

The standardized structure of DwC-A and CF-NetCDF promotes interoperability, allowing data to be shared and integrated across scientific domains. POLARIN leverages these strengths to bring together heterogeneous datasets, creating a unified and accessible data ecosystem that supports and aims to advance interdisciplinary research collaboration.

By adopting DwC-A and CF-NetCDF, POLARIN complies with ENVRI-FAIR recommendations[3], which emphasize the importance of using internationally recognized standards, machine-actionable metadata and interoperability to achieve FAIR data principles.

NOTE When it is not possible to encode data as CF-NetCDF or DwC-A, data can be published in a non-proprietary file format that is easy to consume for users (without specific software) accompanied by a detailed product manual (in PDF format).

4.3. Exchange of harvested discovery metadata

NOTE Further details to be clarified.

The POLARIN metadata catalogue will provide the discovery metadata from the harvested external sources through a standard machine-readable endpoint (described as 'mdcat' in Figure 1). This machine-readable endpoint can be accessed by the human front-end described in Section 3.3.

[1] GBIF Darwin Core:https://www.gbif.org/darwin-core

[2] https://obis.org/

[3] For more details, visit https://envri.eu/envri-fair/